Experimental Setup for Encarsia Formosa Wing Mechanism

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Abstract

This project aimed to design, fabricate, and experimentally validate a **dynamically scaled robotic setup** that replicates the **clap-and-fling aerodynamic mechanism** used by very small insects such as *Encarsia formosa*. Operating at extremely low Reynolds numbers (Re ≈ 10), where conventional lift mechanisms become ineffective, these insects exploit specialized wing kinematics to generate sufficient aerodynamic forces. The experimental platform uses a pair of rigid polycarbonate wings driven by NEMA-17 stepper motors, connected through bevel gears and a rack-and-pinion mechanism enabling simultaneous rotation and linear translation. The system operates in a glycerin–water solution to achieve the required viscosity. Since mid-semester, the mechanical assembly has been completed, Arduino Mega-based motion control code has been implemented, and preliminary motion tests have been successfully conducted. This report presents the complete design, fabrication, control implementation, and current experimental status of the project.

1. Problem Definition

- Very small flying insects operate in a low Reynolds number aerodynamic regime (**Re ≈ 10**) where viscous forces dominate inertial forces. In this regime, conventional aerodynamic lift mechanisms become inefficient, and insects such as *Encarsia formosa* rely on **clap-and-fling** wing kinematics to generate sufficient lift.

- During the clap phase, the wings approach each other at the top of the stroke, expelling fluid downward and building circulation. During the fling phase, the wings rotate apart, drawing fluid into the gap and creating strong leading-edge vortices that enhance lift generation.

- Understanding this mechanism is important for:
  - Advancing biological knowledge of micro-insect flight.
  - Informing the design of efficient Micro Air Vehicles (MAVs).
  - Improving computational fluid dynamic models for low Reynolds number flows.
  - The goal of this project is to replicate the clap-and-fling motion using a dynamically scaled robotic model and study the resulting aerodynamic forces.

2. Literature Survey

2.1 Aerodynamics at Low Reynolds Numbers

- Miniature insects operate in flow regimes governed by the Reynolds number $\mathbf{Re = \rho U c/\mu}$, where ρ is fluid density, U is characteristic wing velocity, c is chord length, and μ is dynamic viscosity. As insect size decreases, both U and c decrease, yielding Re values as low as 10. In this regime, viscous forces dominate and conventional pressure-based lift generation is ineffective; insects instead rely primarily on drag-based mechanisms and unsteady vortex effects.

- Ford et al. (2019) demonstrated that tiny flying insects operate at wing-chord based Reynolds numbers on the orders of 1 to 10. At Re below 30, dimensionless lift coefficients of a revolving elliptical wing increase slightly compared to higher Re, but dimensionless drag coefficients **increase by several hundred percent**. This greatly increases the energetic demands on tiny insects, which must continuously flap to remain aloft.

2.2 Clap-and-Fling Mechanism

- The clap-and-fling motion, first described by **Weis-Fogh (1973)**, is one of the most important aerodynamic mechanisms in miniature insect flight and has since been extensively studied for insects such as *Encarsia formosa*. The mechanism was observed in *Encarsia formosa*, Muscidifurax raptor, and Nasonia vitripennis (Miller and Peskin, 2009).

- Experimental and computational studies confirm that clap-and-fling kinematics provides aerodynamic benefits via:
- Generating bound circulation at the leading edges of the wings during fling with little to no circulation at the trailing edges, conducive for lift generation.
- Generating downward flow during clap that can be used to generate additional thrust for maneuvering.
- More lift enhancement at low Re relevant to tiny insect flight, as compared to larger Re where viscous forces are lower.

2.3 Bristled vs. Solid Wings Findings from Ford et al. (2019)

- The primary reference for this project, **Ford et al. (2019)**, published in *Bioinspiration & Biomimetics*, presented a robotic platform replicating clap-and-fling kinematics in a high-viscosity fluid. Their study examined the aerodynamic effects of varying the ratio of membrane area (AM) to total wing area (AT) in bristled wings at Re ≈ 10.

- Key findings relevant to this project include:
- Bristled wings with AM/AT in the range of 15%–30% (matching thrips forewings) generated the largest lift-to-drag ratios during clap and fling at Re = 10.
- Bristled wings preserved the vast majority of lift achieved by solid wings while substantially reducing drag, due to leaky inter-bristle flow.
- Kolomenskiy et al. showed that bristled wings produce 66–96% of the aerodynamic drag of solid membrane wings, confirming viscous effects dominate at these scales.
- The clap-and-fling mechanism becomes less beneficial at Re = 120, regardless of the drag reduction provided by bristled wings.

- This project uses fully membraned polycarbonate wings to isolate membrane-wing aerodynamics and compare with the bristled wing results of Ford et al. (2019).

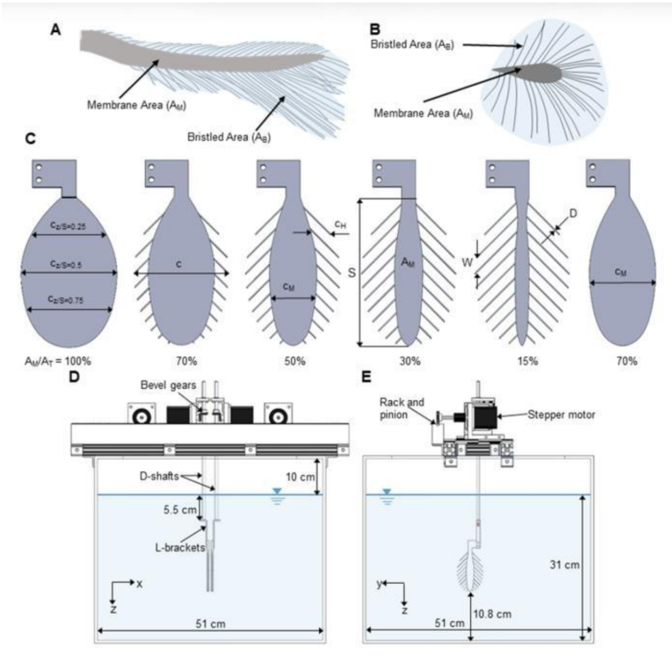


Figure 2.1: Wing models and robotic platform from Ford et al. (2019). Bristled wing models with AM/AT ratios from 15% to 100% (solid wing) are shown, along with the dynamically scaled robotic clap-and-fling platform including stepper motors, rack and pinion mechanism,

bevel gears, and D-shafts. This platform serves as the primary reference for the present experimental design.

2.4 Dynamically Scaled Experiments

- Dynamically scaled robotic wing experiments where wing models are enlarged, motion slowed, and fluid viscosity increased are the standard approach for studying insect aerodynamics at the correct Reynolds number. The robotic model used by Ford et al. (2019) used the same kinematic profile adopted in this project: a 50% overlap between rotation and translation during fling, and 100% overlap during clap. This approach is well-validated against published force data.

3. Experimental Design Parameters

- The experimental setup is dynamically scaled to match the aerodynamic conditions of Encarsia formosa. The key parameters are summarised below:

Parameter	Value / Description
Wing span	90 mm
Mean chord	45 mm (aspect ratio = 2)
Wing thickness	2.5 mm (polycarbonate)
Root spacing	~40 mm
Initial gap (at stroke start)	~4–5 mm (≈ 0.1c)
Wing translation velocity	~0.19 m/s
Fluid medium	Glycerin–water solution
Kinematic viscosity (ν)	≈ 860 mm²/s
Target Reynolds number	Re ≈ 10
Tank dimensions	500 × 300 × 300 mm (approx.)
Stroke period	~2 seconds per full cycle

- The Reynolds number condition is maintained using **Re = (U × c) / ν**. With a wing translation velocity of approximately 0.19 m/s and glycerin–water kinematic viscosity of ≈ 860 mm²/s, the experiment operates at Re ≈ 10 representative of the actual insect. This matches the experimental conditions of Ford et al. (2019), enabling direct comparison of measured forces.

4. Mechanical Design

4.1 Structural Frame

- A rigid metal frame mounted on top of the glass tank supports the motors, wing shafts, and translation mechanism. The frame maintains precise alignment between the two wings throughout the motion cycle and is designed to minimise vibration during operation.

4.2 Transmission System

- Each wing is mounted on a D-profile shaft serving as its rotational axis. The transmission system consists of:
- A 90° bevel gear pair (1:1 ratio) transmitting stepper motor rotation to the wing shaft.
- A pinion gear (module 2, 18 teeth, pitch diameter ≈ 30 mm) on the same shaft engaging with a linear rack, converting rotation into linear translation of the wing assembly.
- MGN15 linear guide rails ensuring smooth, low-friction motion throughout the stroke.

4.3 Motion Profile

- Each full cycle consists of two phases:
- Fling phase: Wings rotate apart (±45°) while translating outward by ~45 mm.
- Clap phase: Wings rotate toward each other while translating inward simultaneously.

- A typical stroke period of approximately 2 seconds maintains the required velocity and Reynolds number conditions. The pinion circumference of ~94 mm means each full stroke corresponds to less than half a revolution of the pinion. This kinematic profile matches the motion prescribed in Ford et al. (2019), where a **50% overlap between rotation and translation** was prescribed during fling, and a **100% overlap** during clap.

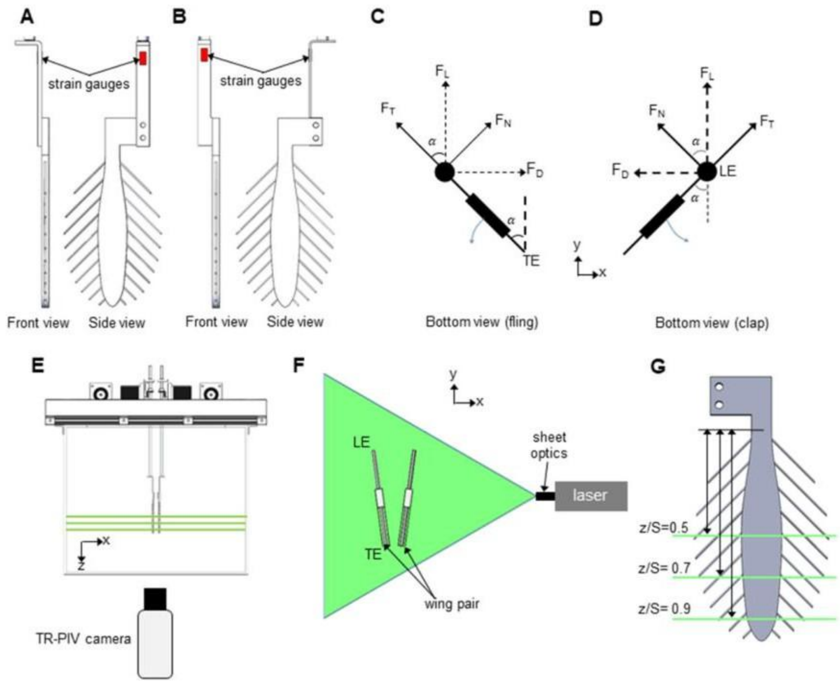


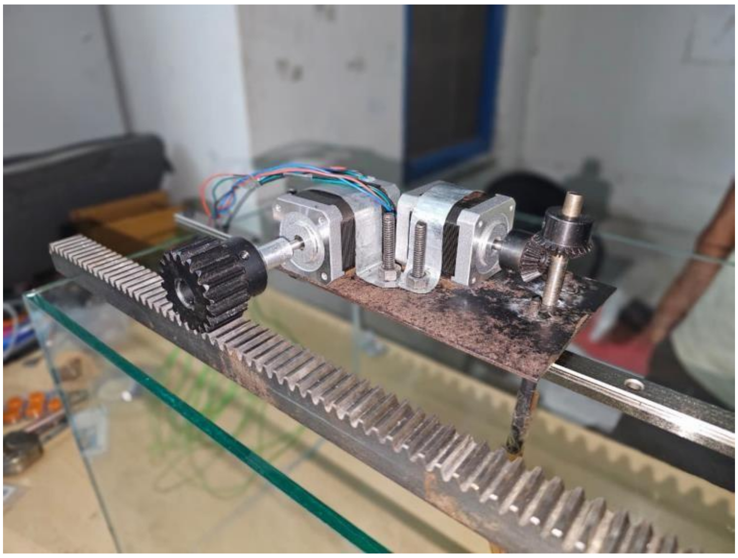
Figure 4.1: Clap-and-fling kinematics from Ford et al. (2019). (A) Matchstick diagram of fling phase showing wing positions at $\tau = 0, 0.2$, and 0.4. (B) Matchstick diagram of clap phase ($\tau = 0.5, 0.8, 1.0$). (C) Time-varying motion profile: rotation angle (thick line) and non-dimensionalised translation distance (thin line). The shaded region ($\tau = 0.8$–1.2) represents the clap-and-fling interaction window where force data are acquired. The present project replicates this kinematic profile.

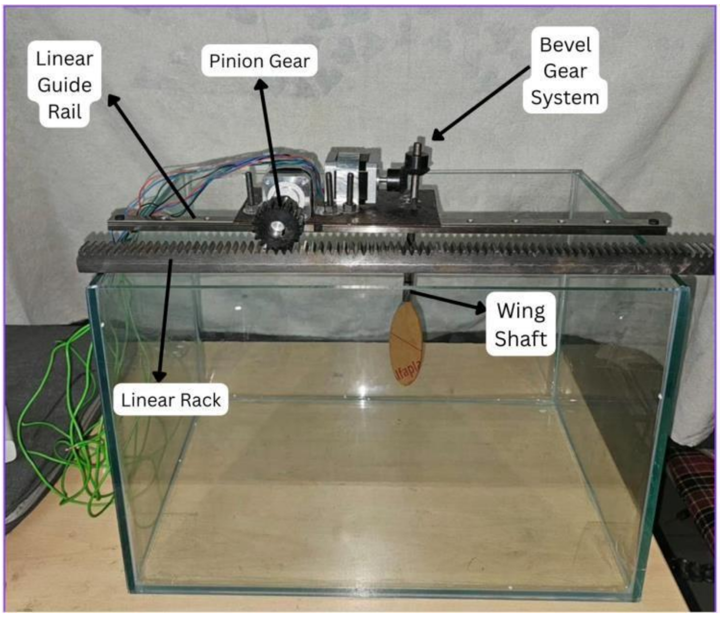
4.4 Torque and Velocity

- The required translation velocity of 0.19 m/s corresponds to a pinion speed of approximately 100 RPM. The aerodynamic drag on the wing is estimated at ~0.3 N, giving a torque requirement of ~0.1 N·m at the pinion. NEMA-17 stepper motors (holding torque ≈ 0.5 N·m) provide a comfortable safety factor of 5×.

4.5. Fabrication and Assembly

- Following the mid-semester design phase, all required components were procured and the mechanical assembly was completed. Key fabrication steps included:
- The metal frame was fabricated and mounted on top of the glass tank to carry the motor assemblies and rack-and-pinion rails.
- NEMA-17 stepper motors were mounted on the frame with bevel gear pairs connecting their shafts to the wing D-shafts at 90°.
- Pinion gears were press-fitted onto the wing shafts and meshed with the linear steel racks.
- The rack-and-carriage assembly was integrated with MGN12 linear guide rails for smooth translation.
- A wooden foil-shaped test piece was attached to the wing shaft to serve as a surrogate wing for initial dry testing before the polycarbonate wings were fitted.
- The current prototype has been assembled and verified for mechanical motion. Photographs of the fabricated setup illustrate the gear train, rack-and-pinion mechanism, and the complete assembly on the tank.





5. Calculations

5.1 Experimental Design Parameters

The design of the robotic model is based on dynamically scaled parameters taken from previous studies.

Wing Geometry: Span = 90 mm, Mean Chord Length= 45 mm Thickness = 2.5 mm polycarbonate

Wing Separation: 500 × 500 × 300 mm transparent acrylic tank

Currently Working Fluid: Water

Target Kinematic Viscosity: $\nu \approx 1$ mm2/s

Reynolds Number Calculation: $Re = (U \times c) / \nu$ where; U = wing velocity, c = wing chord length, ν = kinematic viscosity

For U = 0.19 m/s, c = 0.045 m, ν = 0.0001 mm2/s

$Re \approx 8550$

5.2 Transmission System

The transmission system consists of three main elements: Stepper Motor, Bevel Gear Pair Rack and Pinion Mechanism

Pinion Specifications: Module = 2 mm, Number of Teeth = 18, Pitch Diameter = 36 mm

Pinion Circumference $C = \pi \times D \approx 113.04$ mm

This means that one full rotation of the pinion translates the rack by approximately 113 mm.

5.3 Velocity and RPM Calculations

Desired translation velocity: $U = 0.19$ m/s

Pinion circumference: $C = 0.113$ m

Required rotations per second: rev/s = $U / C \approx 1.681$

Stroke length = 45 mm

Number of revolutions required:

Revolutions = Stroke / Circumference = 45 / 113 = 0.39 revolutions

The wings rotate approximately 45° during the fling phase.

Convert degrees to revolutions: 45° = 45 / 360 = 0.125 revolutions

The wings rotate approximately 45° during the clap phase.

5.4 Torque Calculation

Estimated aerodynamic drag force on wing: $F \approx 0.3$ N (From research Paper)

Pinion radius: $r = 18$ mm = 0.018 m

Torque required: $T = F \times r = 0.3 \times 0.015 = 0.0054$ Nm

Applying safety factor of 10: Required torque ≈ 0.054 Nm

Selected Motor: NEMA-17 stepper motor which has holding torque ≈ 0.7 Nm

This provides more than ten times the required torque ensuring reliable operation.

5.5 Motion Profile

The clap-and-fling cycle consists of two phases:

Fling Phase: The wings rotate apart while translating outward.

Rotation and translation overlap by 50%.

Clap Phase: The wings translate inward and rotate together. Rotation and translation overlap by 100%. Total rotation amplitude: ±45°

Typical stroke period: $T \approx 2$ seconds

Translation per stroke: ≈ 45 mm

5.6 Gear System Design Calculations

Rack and Pinion Selection: The rack and pinion mechanism convert motor rotation into linear translation of the wing assembly. A module 2 pinion gear with 18 teeth is selected because it is commonly available and provides sufficient strength.

Pitch diameter of pinion gear: $D = m \times Z$ where

m = module, Z = number of teeth

$D = 2 \times 18 = 30$ mm

Pinion radius: $r = D/2 = 18$ mm

Translation Speed and Pinion RPM required wing translation velocity $U = 0.19$ m/s

Pinion rotational speed(n) = $U / C = 0.19 / 0.113 = 1.68$ rev/s

Converting to RPM= $1.68 \times 60 \approx 100$

Therefore the pinion must rotate at approximately 100 RPM to achieve the required translation velocity. Torque Requirement for Rack Motion

Bevel Gear Selection

A 90 degree bevel gear pair with a 1:1 ratio is used to transmit motion from the motor shaft to the wing shaft. Gear parameters

Module = 1mm Teeth number = 25

Pitch diameter $D = m \times Z = 1 \times 25 = 25$ mm

Since the gear ratio is 1:1, the bevel gear only changes the direction of rotation and does not change the speed or torque.

5.7 Lift and Drag Force Calculation:

The calculation of aerodynamic forces using flow velocity can be done by:

- We can obtain the detailed wing motion (kinematics) using high-speed cameras. Using this motion data as an input boundary condition, we can numerically solve the incompressible Navier-Stokes equations to determine the flow velocity and pressure fields around the wing Model
- Once we get the flow velocity and pressure fields, the aerodynamic forces (like drag and lift) acting on the wing can be calculated by integrating two main components over the wing's surface:

    - **Pressure Force:** The force resulting from the pressure difference between the windward (front) and leeward (back) surfaces of the wing
    - **Frictional Force (Viscous Stress):** The tangential force caused by the fluid's viscosity as it shears against the surface of the wing.

6. Control System Implementation

6.1 Hardware

- An Arduino Mega microcontroller drives four TB6600 microstepping drivers, one for each NEMA-17 stepper motor. The two motors on each wing control translation and rotation independently, while the Arduino coordinates all four motors simultaneously to produce the correct clap-and-fling kinematics.

6.2 Software AccelStepper Library

- The AccelStepper library was used to enable simultaneous, non-blocking control of all four motors with configurable acceleration and speed profiles. The motion is divided into two phases per cycle:
- Fling phase: Rotation motors drive wings apart (+rotationSteps / −rotationSteps) while translation motors drive wings outward (+translationSteps / −translationSteps) concurrently.
- Clap phase: All four motors reverse direction, bringing wings back together simultaneously.

6.3 Motion Control Code

```
#include <AccelStepper.h>

// Motor definitions (STEP pin, DIR pin)

AccelStepper motor1(AccelStepper::DRIVER, 2, 3); // Translation left AccelStepper
motor2(AccelStepper::DRIVER, 4, 5); // Translation right AccelStepper
motor3(AccelStepper::DRIVER, 6, 7); // Rotation left AccelStepper motor4(AccelStepper::DRIVER, 8,
9); // Rotation right

    •

int translationSteps = 750; // ~45 mm stroke int rotationSteps  =
200; // ~45 deg rotation void setup() {
    motor1.setMaxSpeed(1000); motor2.setMaxSpeed(1000); motor3.setMaxSpeed(800);
    motor4.setMaxSpeed(800); motor1.setAcceleration(500); motor2.setAcceleration(500);
    motor3.setAcceleration(400); motor4.setAcceleration(400);
}
void loop() {
    // -- Fling phase --
    motor3.move(rotationSteps);  motor4.move(-rotationSteps); motor1.move(translationSteps);
    motor2.move(-translationSteps); runAllMotors();  delay(300);
```

```
// -- Clap phase --
motor1.move(-translationSteps); motor2.move(translationSteps); motor3.move(-
rotationSteps); motor4.move(rotationSteps); runAllMotors();   delay(500); }
```

The current code is configured for a single-wing prototype. In the dual-wing setup, motor2 and motor4 will be activated symmetrically. Step counts and speeds will be calibrated against measured translations and angular displacements.

7. Current Experimental Status

- As of the end-semester stage, the following milestones have been achieved:
- Full mechanical assembly of the rack-and-pinion, bevel gear, and frame system completed and mounted on the glass tank.
- Single-wing prototype assembled and tested for mechanical motion integrity.
- Arduino Mega control code implemented and verified all four motors respond correctly to commanded step profiles.
- Dry-run motion tests in air performed to confirm correct fling and clap kinematics without mechanical binding or excessive vibration.
- Preliminary water tests initiated with the single-wing prototype to observe fluid interaction before transitioning to the glycerin–water solution.
- Remaining work includes fitting the final polycarbonate wings, preparing and filling the tank with the glycerin–water mixture to achieve $\nu \approx 860$ mm²/s, and performing force measurement experiments.

8. Future Work

- The immediate next steps for completing the experimental validation are:
- Fabricate and install precision-cut polycarbonate wings (90 mm span × 45 mm chord, 1–1.5 mm thick).
- Prepare the glycerin–water mixture and fill the tank to achieve the target kinematic viscosity of ≈ 860 mm²/s.
- Extend the control code to coordinate both wings symmetrically for full clap-and-fling experiments.
- Calibrate step counts against measured displacements.
- Integrate force sensors (load cells or strain gauges on L-brackets, following the approach of Ford et al. 2019) to measure aerodynamic drag and lift forces on the wings during operation.
- Run parametric experiments varying stroke frequency and wing gap to study their effects on force generation.
- Compare measured force coefficients with published data from Ford et al. (2019) and Kolomenskiy et al. to validate the experimental model.
- In future iterations, explore bristled polycarbonate wings with AM/AT ≈ 15%–30% to replicate the biological configuration of Encarsia formosa.

9. Conclusion

- This project has successfully progressed from concept through design, fabrication, and initial testing of a dynamically scaled robotic wing apparatus that replicates the clap-and-fling aerodynamic mechanism of Encarsia formosa. The mechanical system comprising NEMA-

17 stepper motors, 90° bevel gears, and a rack-and-pinion translation mechanism produces the simultaneous rotation and translation required for clap-and-fling kinematics. The Arduino Mega control system, implemented using the AccelStepper library, successfully coordinates all four motors to execute the correct motion profile.

- The experimental design closely follows the validated methodology of Ford et al. (2019), including identical wing dimensions (90 mm span, 45 mm chord), the same glycerin–water kinematic viscosity (860 mm^2/s), and an equivalent kinematic overlap profile. This ensures that the force measurements to be performed in the next phase can be directly benchmarked against the published reference data.

- Preliminary testing confirms that the mechanism operates smoothly and produces the intended kinematics. The next phase will involve full glycerin-solution testing with polycarbonate wings and aerodynamic force measurements. The completed platform will contribute to the understanding of low Reynolds number flight and provide a validated experimental basis for future bio-inspired MAV research.

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- Additionally, the project has provided valuable practical exposure in interdisciplinary engineering areas including mechanical design, embedded control systems, motion synchronization, and experimental fluid mechanics. The integration of fabrication, electronics, and aerodynamic principles into a single working platform demonstrates the effectiveness of the overall system design and establishes a strong foundation for further experimental investigations.

10. References

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